

Non-Linear Data Structures: Graph

گراف لچکدار روابط کے لیے ہے: کوئی سخت درجہ بندی نہیں، کوئی بھی نوڈ کسی سے جڑ سکتا ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 20 of 24

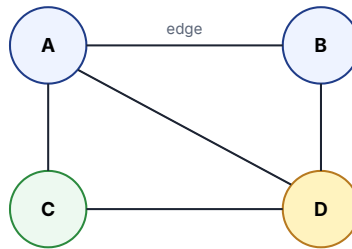
Non-Linear Data Structures: Graph

LEARN IT

A **graph** organises data when things connect in a flexible, complex way. Unlike a tree, there is **no strict hierarchy**: any node can connect to any other node.

- **Nodes (vertices)** — the items (cities, people).
- **Edges** — the connections (roads, friendships).

Examples: Facebook friends; GPS mapping (the fastest route between two cities).



Four vertices A, B, C, D. Edges can form a cycle (A–B–D–C–A). A tree cannot do that.

WORKED EXAMPLE

Worked example: Karachi – Hyderabad – Sukkur – Larkana. Cities are vertices. Roads are edges. You can travel Karachi → Hyderabad → Sukkur or Karachi → Larkana → Sukkur. There is no single “root city”. That is a graph, and GPS uses it to pick a fast route.

REMEMBER IT

Tree $\subset$ Graph in shape-talk: a tree is a special graph with one root and no cycles. If you need cycles or many-to-many links, say **graph**.

COMMON MISTAKE

Mistake: using a tree for GPS because “Hyderabad comes under Sindh”.

Fix: administration can be a tree; **roads** are a graph because you can loop and take alternative paths.

اردو میں سمجھیں

گراف لچکدار روابط کے لیے ہے: کوئی سخت درجہ بندی نہیں، کوئی بھی نوڈ کسی سے جڑ سکتا ہے۔ نوڈز (Vertices) چیزیں ہیں، کنارے (Edges) تعلق۔ فیس بک دوست اور GPS نقشہ کلاسیکی مثالیں ہیں۔ چکر (Cycle) اور کئی راستے گراف کو ٹری سے الگ کرتے ہیں۔

CHECK YOURSELF

1. What are the two parts of a graph?
2. Give two source examples of graphs.
3. State one difference between a tree and a graph.

Answers: 1. Nodes/vertices and edges. 2. Social networks (Facebook friends) and GPS mapping. 3. A tree has a hierarchy and one root; a graph has no strict hierarchy and any node may connect to any other (cycles allowed).

Next lecture: 2.5.3 Operations on Data Structures